

Ezau Faridh Torres Torres

📍 San Luis Potosí, SLP., México ✉ ezau.torres@ciimat.mx ☎ (+52) 487 124 25 39 in ezautorres 📧 ezautorres

Applied mathematician (MSc @ CIMAT) specialized in NLP, deep learning, data science, and statistical modeling. Experienced in real-world text analytics and scientific computing, with a strong foundation in math and Machine Learning.

Education

Center for Research in Mathematics, A.C. (CIMAT) [🔗](#)

Aug 2023 - June 2025

MSc in Applied Mathematics. GPA: 9.7/10.0

Thesis: Physics-Informed Neural Networks for Bayesian Inverse Problems

Autonomous University of San Luis Potosí (UASLP) [🔗](#)

Aug 2017 - Dec 2021

BSc in Applied Mathematics. GPA: 9.1/10.0

Experience

Research assistant [🔗](#)

Guanajuato, Gto.

Center for Research in Mathematics, A.C. (CIMAT) [🔗](#)

Aug 2024 - Jul 2025

- Contributed to a research project on Bayesian Uncertainty Quantification developing surrogate models with PINNs in PyTorch and employed probabilistic models for uncertainty analysis. [🔗](#)
- Co-authored a paper in preparation on deep Bayesian surrogate modeling applied to scientific simulations.

Research assistant

San Luis Potosí, SLP.

Potosino Institute of Scientific and Technological Research, A.C. (IPICYT) [🔗](#)

Mar 2022 - Jan 2023

- Contributed to a research project analyzing mathematical models of neuronal responses to external stimuli. Conducted simulations and sampling of probability distributions modeling neuron action rates and applied advanced mathematical techniques to improve the understanding of neuronal dynamics.

Technical Proficiencies

Programming Languages: Python, R, MATLAB, SQL, Git, Wolfram Mathematica, wxMaxima, C++.

Technologies & Tools: PyTorch, TensorFlow, Keras, Scikit-learn, Jupyter, Pandas, NumPy, PowerBI, HuggingFace, transformers, fastText, GitHub, Google Cloud, Pyspark, SciPy, Matplotlib, DBeaver, Seaborn, LaTeX, Linux.

Languages: English IELTS C1.

Relevant Coursework

- **Deep Learning & Machine Learning:** Natural Language Processing [🔗](#), LLMs, IA & Advanced Deep Learning [🔗](#), Big Data, Pattern Recognition, Machine Learning techniques, Optimization [🔗](#), Scientific Computing for Data Science [🔗](#), Computer Vision [🔗](#), Data Preparation, Data Manipulation, Model Optimization, Predictive Modeling, Algorithms.
- **Mathematics & Modeling:** Mathematical Modeling, Advanced Numerical Methods [🔗](#), Numerical Analysis, Dynamical Modeling, Analytical Modeling, Analytical Stochastic Processes, Matrix Methods for Data Analysis.
- **Statistics & Analysis:** Probabilistic and Statistical Models, Statistical Analysis, Bayesian Inference, Uncertainty Quantification (MSc Thesis topic) [🔗](#), Markov Chain Monte Carlo Methods [🔗](#).
- **Other Technical Areas:** Geometry and Computer-Aided Design, Optimal Control in Robotics (BSc Thesis topic).

Portfolio Projects

Rest-Mex 2025 - NLP Challenge

[Github](#) [🔗](#)

- Multitask classification of Spanish-language tourist reviews using Hierarchical Attention Networks (HAN).
- Ranked in the top 10% using Hierarchical Attention Networks (HAN) to classify sentiment, destination type, and Magic Town from Spanish-language reviews.

PINNs Framework

[Github](#) [🔗](#)

- Modular solver for PDEs with Physics-Informed Neural Networks.
- Built a PyTorch-based framework to solve forward and inverse PDEs in several domains, incorporating uncertainty quantification with MCMC.

Fraud Detection Model API

[Github](#) [🔗](#)

- Fraud Detection API - End-to-end pipeline for insurance fraud detection with WOE + logistic regression, deployed via FastAPI for real-time scoring.

Multitask Tweet Classification

[Github](#) [🔗](#)

- Joint prediction of gender and nationality from Spanish tweets.
- Combined RoBERTa embeddings and TF-IDF features in a hybrid model trained with multitask loss and evaluated via joint accuracy.

Relevant Courses & Certificates

- Google Advanced Data Analytics Professional Certificate — Google Career Certificates.
- Data Science School (UNAM) · Industrial Problem-Solving Workshop (CIMAT) · Numerical Modeling School (CIMAT)